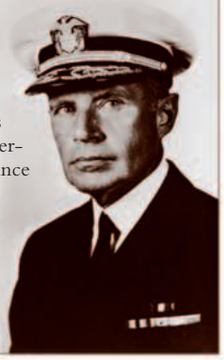
1886–1969

RAYMOND A SPRUANCE

US ADMIRAL AND COMMANDER OF US FORCES AT MIDWAY AND THE PHILIPPINE SEA

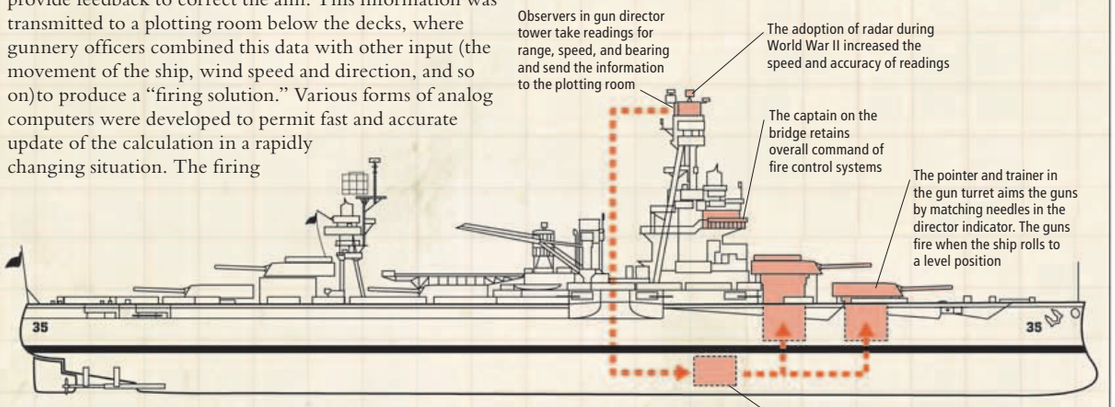
A career naval officer, Raymond Spruance commanded a cruiser division in the early stages of the Pacific War. Despite having no previous experience of carriers, he was given command of Task Force 16 in time to play a vital part in the crucial battle of Midway in June 1942. As commander of the US 5th Fleet from 1943 he spearheaded the drive through the central Pacific, presiding over the destruction of Japanese naval aviation at the battle of the Philippine Sea. Sometimes criticized as over-cautious, Spruance enjoyed an exceptional record of success in major naval engagements.



WEAPONS AND TECHNOLOGY

FIRE CONTROL SYSTEMS

The increasing range of naval guns and speed of warships from the late 19th century onward led to the introduction of fire control systems to replace gunners sighting by eye. Personnel aloft in the director tower were equipped with optical instruments to gauge the bearing and range of a target to a fine degree of accuracy. They also "spotted" the splash of shells to provide feedback to correct the aim. This information was transmitted to a plotting room below the decks, where gunnery officers combined this data with other input (the movement of the ship, wind speed and direction, and so on) to produce a "firing solution." Various forms of analog computers were developed to permit fast and accurate update of the calculation in a rapidly changing situation. The firing solution was transmitted electronically to the gun turrets, appearing as a visual pointer indicating the correct elevation and bearing. The gunner matched the pointer on his gun to the desired position and the guns were ready to fire. The use of radar and increasingly complex computers allowed large strides to be made in speed and accuracy of fire during World War II.



Director firing
Most of the guns on a large warship were directed—the person firing the gun relied on a series of readings and calculations taken elsewhere on the ship.



Wrecked Japanese transport
Japanese cargo ship *Kinugawa Maru* lies beached and gutted on the Guadalcanal coast, destroyed by US aircraft after the naval battle of Guadalcanal.

WORLD WAR II

TASSAFARONGA



Date November 30, 1942
Forces Americans: 5 cruisers, 4 destroyers; Japanese 8 destroyers
Losses Americans: 5 cruisers, 4 destroyers; Japanese: 8 destroyers
Location Off Guadalcanal, Solomon Islands

In response to a shortage of supplies, Japanese destroyers under Rear Admiral Raizo were delivering drums of food by night to troops based on Guadalcanal when they were intercepted by Task Force 67 commanded by Rear Admiral Wright. The five cruisers in Wright's force had an overwhelming superiority in firepower and possessed radar, which the Japanese did not. On detecting the Japanese ships there was a brief delay before permission to commence firing was granted. However, once the word was given, the Americans concentrated their fire on a single ship, *Takanami*. She was quickly incapacitated, but in the interim the other Japanese destroyers had converged. None of the torpedoes fired by the American force found their mark. Guided onto target by the flashes of the American guns, the Japanese destroyers delivered a series of devastating torpedo attacks that left *Minneapolis*, *New Orleans*, and *Pensacola* holed and on fire. *Northampton* was the last cruiser hit; struck by two torpedoes, she sank in the course of the night. All of the Japanese destroyers except *Takanami* escaped unscathed, but did not succeed in delivering the supplies.